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Nrf2 Regulates Neurogenesis and Protects Neural Progenitor Cells Against Aβ Toxicity

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ABSTRACT

Neural stem/progenitor cells (NPCs) proliferate and produce new neurons in neurogenic areas throughout the lifetime. While these cells represent potential therapeutic treatment of neurodegenerative diseases, regulation of neurogenesis is not completely understood. We show that deficiency of nuclear factor erythroid 2-related factor (Nrf2), a transcription factor induced in response to oxidative stress, prevents the ischemia-induced increase in newborn neurons in the subgranular zone of the dentate gyrus. Consistent with this finding, the growth of NPC neurospheres was increased by lentivirus-mediated over-expression of Nrf2 gene or by treatment with pyrrolidine dithiocarbamate (PDTC), an Nrf2 activating compound. Also, neuronal differentiation of NPCs was increased by Nrf2 over-expression or PDTC treatment but reduced by Nrf2 deficiency. To investigate the impact of Nrf2 on NPCs in Alzheimer's disease (AD), we treated NPCs with amyloid beta (Aβ), a toxic peptide associated with neurodegeneration and cognitive abnormalities in AD. We found that $A\beta_{1-42}$-induced toxicity and reduction in neurosphere proliferation were prevented by Nrf2 over-expression, while Nrf2 deficiency enhanced the $A\beta_{1-42}$-induced reduction of neuronal differentiation. On the other hand, $A\beta_{1-40}$ had no effect on neurosphere proliferation in wt NPCs, but increased the proliferation of Nrf2 over-expressing neurospheres and reduced it in Nrf2 deficient neurospheres. These results suggest that Nrf2 is essential for neuronal differentiation of NPCs, regulates injury-induced neurogenesis and provides protection against Aβ-induced NPC toxicity. STEM CELLS *2014; 00:000–000*

INTRODUCTION

Impaired neurogenesis in aging brain and insufficient induction of neurogenesis have been associated with defective motor or cognitive functions in various brain disorders, including neurodegenerative diseases. Neural stem/progenitor cells (NPCs) have thus high therapeutic potential for a large number of indications. There are two alternatives to repair or restore the function of the diseased brain by taking advantage of NPCs: to enhance or protect endogenous neurogenesis, or to transplant exogenous NPCs [1]. Endogenous NPCs localize to two main restricted regions: the subventricular zone and subgranular zone (SGZ) of the hippocampal dentate gyrus (DG). Here, the NPCs continuously produce new neurons [2], but this neurogenesis is severely impaired during aging. Importantly, even though neurogenesis is triggered after brain insults, these spontaneously born

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NPCs appear not to be sufficient for significant recovery. Moreover, neurogenesis is further impaired in psychiatric diseases, such as depression [3,4], or neurodegenerative diseases, such as Alzheimer's disease (AD) [5], thereby possibly contributing to the memory loss in AD brain [6].

One key mechanism behind impaired neurogenesis in neurological diseases, including AD is increased oxidative stress which has been shown to interfere with neural proliferation, differentiation and survival. In AD in particular, the main pathological characteristics are extracellular deposits of amyloid beta (Aβ) and intracellular accumulation of neurofibrillary tangles containing a hyperphosphorylated microtubule-associated protein, tau [7]. The amyloid hypothesis suggests that Aβ is responsible for the pathological and cognitive abnormalities in AD [8-9]. Increased oxidative stress associated with Aβ toxicity together with the secondary inflammation may have negative effects also on the transplanted NPCs impairing their therapeutic effectiveness. Thus, enhanced defense potential of the transplanted cells against oxidative stress could increase survival of the transplanted cells and also relieve oxidative stress in the area surrounding transplanted NPCs.

An endogenous cellular defense mechanism against oxidative stress is induction of the production of endogenous antioxidant enzymes and protective proteins at the transcriptional level [10]. Over 100 cytoprotective genes share a common promoter element called the antioxidant response element (ARE). Nrf2 is a transcription factor that binds to the ARE and promotes the transcription of these protective genes [11]. Nrf2 is expressed in a constitutive manner and found both in the cytoplasm as well as in the nucleus. Under normal conditions kelch-like ECH-associated protein 1 (Keap 1) binds Nrf2 and retains it in the cytoplasm where it is degraded by the proteasome [12]. Under conditions of oxidative stress, Nrf2 degradation is reduced, leading to its accumulation, translocation into the nucleus and induction of various phase II detoxification enzymes including glutathione-S-transferase (GST), heme oxygenase-1 (HO-1), NAD(P)H:quinone oxidoreductase 1 (NQO1), γ-glutamatecysteine ligase catalytic (GCLC) and modifier (GCLM) subunits[10,13-15]. Besides inducing protective gene expression, Nrf2 can directly inhibit Fas-mediated apoptosis [15-16] and induce the expression of a class of proteasomal proteins, thus reducing protein aggregation [17-18]. Moreover, Nrf2 can also attenuate the inflammatory response [19-20]. Considering the large role of Nrf2 in mechanisms relevant to brain injury and neurodegeneration, it is not surprising that basic research on Nrf2 has suggested a potentially therapeutic role of Nrf2-ARE pathway both in acute injuries of the central nervous system (such as stroke, intracerebral hemorrhage, traumatic brain injury and spinal cord injury)[21-23] and neurodegenerative diseases (such as Alzheimer's disease (AD), Parkinson's disease, Huntington's disease, multiple sclerosis and amyotrophic lateral sclerosis) [24-26]. Importantly, in 2009 Zhao and coworkers have described a novel role for Nrf2: it promotes neuronal differentiation in the SH-SY5Y neuroblastoma cell line, further adding to the beneficial attributes of this transcription factor [27]. Moreover, Nrf2 has been shown to mediate minocycline-induced precondition of neural stem cells in ischemic conditions [21].

Given the important role of oxidative stress in AD, a promising approach might be to reduce reactive oxygen species (ROS)-related neuronal injury by modulating Nrf2 pharmacologically or by stimulating Nrf2 overexpression [28]. Potential Nrf2 inducers include sulphoraphane, pyrrolidine dithiocarbamate (PDTC) and tert-butylhydroquinone (tBHQ) [29-32]. We and others have shown that Nrf2 is protective against Aβ toxicity *in vitro* [33-34] and that gene transfer of Nrf2 into the hippocampi of transgenic AD mice improves spatial learning [35]. Nrf2 also reduces oxidative stress-induced Aβ production [30], and induces the proteosome to remove toxic Aβ [18]. Taken together, these studies highlight the potential therapeutic benefit of induction of the Nrf2 pathway in AD.

While stabilization of Nrf2 by tBHQ has been shown to protect NPCs against oxidative stress-induced cell death [28, 33], the possible role of Nrf2 in NPC proliferation, migration and maturation, as well as in Aβ toxicity is not well characterized. In this report we used $Nrf2^{-/-}$ mice and lentiviral gene transfer to assess the role of Nrf2 in the physiology of NPCs. We showed that Nrf2 is essential for injury-induced neurogenesis in the hippocampus, Nrf2 regulates proliferation and neuronal differentiation but not migration of NPCs, and that Nrf2 protects NPCs against Aβ toxicity.

Materials and Methods

Animal models

All animal experiments were conducted according to the national regulations of the usage and welfare of laboratory animals and approved by the Animal Experiment Committee in State Provincial Office of Southern Finland. For the animal experiments, age- and weight – matched adult male wt (n=13) and $Nrf2^{-/-}$(n=13) [36] mice on a C57BL/6J background were housed in groups (two-three mice per cage) on a 12-h light/dark cycle and given food and water *ad libitum* throughout the study. The mice were anesthetized with 3 % halothane for induction and 1.5 % halothane for maintenance (70% N_2O/30% O_2). Rectal temperature was maintained at 36.5 ± 0.5˚C using a thermostatically controlled rectal probe connected to a homoeothermic blanket during surgery (PanLab, Harvard Apparatus, Spain). A midline cervical incision of approximately 5mm was made starting right above the upper thorax and both common carotid arteries were exposed without damaging the vagus nerve. Both arteries were then enclosed with suture thread, lifted and compressed with microaneurysm clips for 17 minutes. Reperfusion was

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restored by removing the clips and incision sutured. The animals were then placed in an incubator (31˚C) for 24 hours before returning to their home cage. Sham mice underwent a similar surgery but their arteries were not compressed with microaneurysm clips. Seven days after global ischemia the mice were terminally anesthetized with tribromoethanol (Avertin, Sigma-Aldrich, St. Louis, Missouri, USA) and transcardially perfused with heparinized saline (2,500 IU/L) for 3 min before removing the brain. The brains were post fixed by immersion in 4 % paraformaldehyde (PFA) solution for 21 h at 4 ^{o}C, cryoprotected in 30 % sucrose for 48 hours, frozen in liquid nitrogen and stored at –70 ^{o}C.

Immunohistochemistry

For immunohistochemistry 20-µm-thick coronal sections were cut with a cryostat (Leica Microsystems GmH). Proliferation and neurogenesis were assessed by immunohistochemical analysis of Ki-67 and doublecortin (DCX), respectively. Because Ki-67 is an endogenous and specific antigen expressed during all active mitotic phases of the cells, is 50% more sensitive than BrdU in detecting proliferating cells in the dentage gyrus, and is not associated with toxicity, we chose Ki-67 labeling instead of BrdU [37-39], especially when using the time point of 7 days after ischemia when neurogenesis is known to be peaking [40]. DCX is a 40-kDa microtubule-associated protein specifically expressed in neuronal precursors of the developing and adult CNS. Due to its specific expression pattern DCX is seen as a marker for neuronal precursors and neurogenesis [37, 39, 41]. Without double-labeling with proliferation markers, DCX immunoreactivity is a marker for newly-born neuronal precursors while not identifying their recent or ongoing proliferation. To detect the extent of neuronal death, immunostaining of NeuN was used. Coronal sections were first washed with 0.1 M phosphate buffer (PB) for 6 x 30 min, mounted onto slides and air-dried for 20 min. After rehydration the sections were permeabilized in 0.4 % Triton X-100 in PBS for 30 min and washed for 3 x 5 min in PBS with 0.5 % Tween (PBST). Non-specific binding was blocked with 10 % normal goat serum (NGS, Chemicon International) for 1 hour for DCX and Ki-67 or with 0.5% Mouse on Mouse reagent (Vector Laboratories) for NeuN for 1 hour. The sections were incubated overnight with rabbit monoclonal Ki-67 (1:500 dilution, Abcam), rabbit polyclonal DCX antibody (1:200 dilution, Cell Signaling) or mouse monoclonal NeuN (1:200 dilution, Merck Millipore) in 5 % NGS PBST. The sections were then washed in PBST for 3 x 5 min before adding the secondary antibody (goat anti-rabbit Alexa Fluor conjugated 568, 1:200, Molecular Probes or goat anti-mouse Alexa Fluor conjugated 568, 1:200, Molecular Probes) in 5 % NGS PBST at room temperature (RT) for 2 h. Finally, sections were mounted with Vectashed mounting media with DAPI (Vector Laboratories). Four to five sections per animal were imaged with an Olympus AX70 microscope equipped with a digital camera (Color View 12 or F-view; SoftImaging Systems). The DCX, Ki-67 and NeuN immunopositive area was quantified in the hippocampi using Image Pro Plus software (Media Cybernetics).

NPC culture and treatment

NPCs were generated from the hippocampi of Nrf2$^{-/-}$ and wt mice at embryonic day 18 (E18) and cultured as free-floating cell aggregates called neurospheres in the presence of epidermal growth factor (EGF) and basic fibroblastic growth factor (bFGF) as described [42-43]. We chose to use fetal NPCs over adult NPCs as they resemble better the NPCs derived from embryonic stem cells and IPSCs (induced pluripotent stem cells), the sources of possible future transplantation therapies. Moreover, the basic characteristics of fetal NPCs and adult NPCs are similar, yet the availability of adult NPCs from aged mice that would model AD the best is very limited. Neurospheres were differentiated for 7 days with or without the presence of 1 – 10 µM pyrrolidine dithiocarbamate (PDTC, Sigma, P8765). Fresh PDTC was added to the cultures every second day during the experiment. To assess the effect of Aβ on the viability of NPCs $A\beta_{1-42}$ and $A\beta_{1-40}$ (American Peptide) were reconstituted in sterile water at a concentration of 155 µM and used immediately after reconstitution. We have previously performed dose-response assays for Aβ exposure [44] and noticed that the cellular response to Aβ varies slightly from peptide lot to lot, and the concentrations of Aβ used were thus adjusted accordingly. Dead and live cells were detected by probidium iodidepropidiumiodide (PI) and SYTO 13 (Molecular Probes) stainings.

Transduction of cultured NPCs with a lentiviral vector

To overexpress Nrf2 in NPCs, wt neurospheres were dissociated and single NPCs infected with multiplicity of infection (MOI) of 3 at the density of 500 000 cells / ml with a lentiviral vector carrying human Nrf2 (LV-Nrf2) [35] or green fluorescent protein (LV-GFP), produced and titered as in [35]. The MOI of 3 was selected by test experiments which showed that this MOI of LV-GFP resulted in the efficiency of 80% without affecting proliferation while higher MOI had a tendency to alter proliferation of NPCs (data not shown). To ensure that NPCs stay in an undifferentiated state during transduction, the culture media was supplemented with growth factors; 10 ng/ml bFGF and 20 ng/ml EGF (Peprotech, London). To remove transduction media after 18 h, the cells were centrifuged at RT at 167 g for 1 min, resuspended in fresh culture media, and expanded in culture.

Quantitative Real-Time PCR

Quantitative real-time PCR was used for detection of Nrf2 over-expression and expression of NQO1, GCLM, GCLC and HO-1 following of LV-Nrf2 transduction. RNA was extracted using the RiboPure Kit (Ambion). cDNA synthesis was performed using Random hexamer pri-

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mer (Fermentas) and Maxima reverse transcriptase (Fermentas). Relative mRNA expression levels of human Nrf2 were quantified using TaqMan chemistry and Assays-on-demand (Applied Biosystems) according to manufacturer's instructions with ABI PRISM 7700 Sequence detector (Applied Biosystems). Nrf2 expression levels were normalized to ribosomal RNA and presented as fold change in the expression level.

Characterization of NPC phenotype, migration, and proliferation

To characterize the phenotype of differentiating NPCs and their ability to migrate, 10 – 15 neurospheres (diameter of 200 – 400 µm) were collected and plated onto the bottom of poly-DL-ornithine coated plates and grown for 7 days without growth factors. Following differentiation the cells were fixed with 4 % formaldehyde and immunostained. Fixed cells were first permeabilized with ice-cold methanol for 20 min. For BrdU staining of proliferation, the cells were incubated with 2 N HCl for 10 min. After each step of this protocol the cells were washed 3 x 5 min with PBS. Non-specific binding sites were blocked with 20 % NGS in PBS for 1 hour. Primary antibodies were diluted in 5 % NGS in PBS, added to the cells, and incubated overnight at RT. Incubation with the secondary antibody was for 2 hours at RT. For detection of neurons, mouse anti-β III-tubulin (Tuj-1, 1:500 dilution, Biosite Covance) was used followed by the secondary antibody Alexa Fluor 568 goat anti mouse IgG (1:200 dilution, Molecular Probes). For detecting astrocytes, rabbit anti-glial fibrillary acidic protein (GFAP, 1:500 dilution, Z0334, DAKO) was used, and for detecting proliferating cells, mouse anti-BrdU (BrdU, 1:50 dilution, Roche) was used. Nuclei were stained with Hoechst 33342 (2.5 µg/ml, Sigma). For detection of dead and live cells, propidium iodide (PI) (5µM, Molecular Probes) and SYTO 13 (1µM, Molecular Probes) were added to cells and incubated for 10 min at 37 ^{o}C. The cells were imaged immediately after staining and the numbers of live and dead cells were counted. Six to twelve images per group were taken by using Olympus IX71 microscope attached to a DP70 digital camera running DP software (all from Olympus). The length of migration from the edge of neurospheres was measured by using Image Pro Plus software (Media Cybernetics).

To study the proliferation of NPCs, 15 000 single NPCs were plated into 96-well plates in the medium supplemented with EGF and bFGF for 48 hours. During the incubation, NPCs began to proliferate and form new freely floating neurospheres. To evaluate the proliferation of the NPCs, the neurospheres were imaged (6 to 8 images per group) and the average volume (µm3) of the largest (diameter over 40 µm) neurospheres was counted. Proliferation was also studied by using the BrdU pulse-labeling method. Single NPCs were incubated at the density of 200 000 cells/ml in the culture media supplemented with EGF, bFGF and 10 µM BrdU (Sigma) for 4 hours. After incubation cell clusters were dissociated using TryplE, and single NPCs were let to attach onto the poly-DL-ornithine coated 48-well plates overnight. The attached NPCs were fixed with 4 % formaldehyde and processed for BrdU immunostaining as mentioned earlier in this text. Twenty replicate images were taken per wt and LV-Nrf2 group and the numbers of BrdU positive cells were counted.

Confocal Ca^{2+} -imaging

For recording the depolarization of differentiated NPCs, cells were challenged with potassium chloride (KCl). Depolarization induced intracellular calcium $[Ca^{2+}]_i$ transients were detected with an Olympus FV1000 confocal microscope (excitation: 488 nm, detection 500-600 nm). First 10 – 15 mouse LV-Nrf2 and wt neurospheres of average size (200 – 400 µm) were differentiated for 6 days on poly-DL-ornithine coated 35 mm glass bottom petri dishes (Mat Tek corporation) in culture media without growth factors. Before measurement the cells were incubated with 5 µM Fluo-4-AM Ca^{2+} -indicator (Molecular Probes) for 30 min at 37 ^{o}C. For perfusion Na^{+} -Ringer's solution (pH 7.4) consisting of (in mM) 137 NaCl, 5 KCl, 1 $CaCl_2$, 1.2 $MgCl_2$, 0.44 KH_2PO_4, 4.2 $NaHCO_3$, 10 glucose, 1 probenecid and 20 HEPES were used at a perfusion rate of 2 ml/min (37 ^{o}C). Cells were next perfused with a depolarizing solution in which NaCl was iso-osmotically replaced with 70 mM KCl. For recordings only cells which migrated the furthest from the edge of neurospheres were chosen and one microscope field was assessed per each neurosphere. One microscope field consists of 50 – 100 cells. The images were analyzed with the Olympus FluoView software. The fluorescence intensity is expressed as an F/F_0 –ratio, where F is the background subtracted fluorescence intensity and F_0 is the background subtracted minimum fluorescence value measured from each cell at resting state.

Statistical analyses

The data are presented as mean ± standard error of mean (SEM). The data were analyzed with Student´s t-test or one or two way analysis of variance (ANOVA) followed by Bonferroni post-hoc test using SPSS and GraphPad software. The impact of cell genotype on calcium responsiveness was evaluated with Fisher's exact test using GraphPad software. Statistical significance is indicated as *p<0.05, **p<0.01, ***p<0.001.

RESULTS

Nrf2 contributes to endogenous neurogenesis induced by global brain ischemia

Enhanced neurogenesis has been demonstrated to consistently take place in the DG after global ischemia in various species, including mice [21], gerbils [40] and rats [45]. Thus, to study whether Nrf2 is involved in neurogenesis *in vivo* we induced endogenous NPC proliferation in wt and $Nrf2^{-/-}$ mice by subjecting them to global

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brain ischemia (Fig. 1) and stained the hippocampal sections with NeuN, a neuronal marker, as well as with DCX and Ki-67 antibodies, markers of neuronal precursor cells/immature neurons and proliferating cells, respectively. In line with the previous report on focal brain ischemia [46], Nrf2$^{-/-}$ mice developed larger ischemic injury in CA1 pyramidal cell layer 7 days after ischemia [9.78±1.25 (mean ± SEM) for wt and 6.28±0.88 for Nrf2$^{-/-}$ mice, respectively, n=7 per group, $p<0.05$], confirming the increased susceptibility of Nrf2$^{-/-}$ mice to brain ischemia. Regarding neurogenesis, the immunoreactivities for DCX and Ki-67 in Nrf2$^{-/-}$ mouse hippocampal SGZ in sham mice were 83% ± 10% (mean ± SEM) and 91.5% ± 5.7% of the corresponding values of wt mouse hippocampi, respectively, showing thus no statistically significant differences between the two mouse lines without ischemic brain injury (n = 4 per group; $p > 0.05$). Instead, seven days after global ischemia, the SGZ immunoreactivity for DCX in the Nrf2$^{-/-}$ mice was 80% less when compared with wt mice ($p<0.01$) (Fig. 1E). Moreover, induction of Ki-67 staining, a marker of cell proliferation for all brain cells, including NPCs and glia that respond to ischemic injury, also showed a tendency to be reduced in Nrf2$^{-/-}$ mice (by approximately 30%) (Fig. 1F). These results suggest severely impaired induction of neurogenesis in Nrf2$^{-/-}$ mice in response to brain insult.

Nrf2 promotes NPC proliferation and neuronal differentiation *in vitro*

To assess further whether Nrf2 affects the proliferation, differentiation and migration of NPCs, we isolated NPCs from wt mice and Nrf2$^{-/-}$ mice. Nrf2 over-expressing cells were obtained by transducing wt NPCs with a lentiviral vector carrying human Nrf2. The over-expression of Nrf2 in cells transduced with the lentiviral vector was confirmed by qRT-PCR: the expression of human Nrf2 was 400-fold higher when compared to background signal in the non-transduced control group (Fig. 2A). Furthermore, the expression levels of Nrf2-related genes; HO-1, GCLC, GCLM and NQO1 were approximately 1.5-fold higher in Nrf2 over-expressing cells compared to controls (Fig. 2B-E). To examine the effect of Nrf2 on NPC proliferation single NPCs were allowed to proliferate for 48 hours in the presence of growth factors. LV-Nrf2 NPCs formed larger neurospheres in comparison to wt NPCs (Fig. 3A). Quantification of the average volume of neurospheres (diameter over 40 μm) revealed that LV-Nrf2 neurospheres were about 40% larger than wt neurospheres ($p<0.001$), whereas Nrf2–/– neurospheres did not differ significantly from control neurospheres (Fig. 3B). We also studied proliferation by using BrdU pulse labeling method and showed that during the 4 hours BrdU pulse the number of proliferated LV-Nrf2 NPCs was 1.4-fold higher compared to wt NPCs ($p<0.001$) (Fig. 3C-D). Next, the effect of Nrf2 on the migration of NPCs was evaluated. Immediately after plating, single NPCs started to migrate out of the neurospheres, spread around and differentiate into neurons, astrocytes (Fig. 3E) and to lesser extent, oligodendrocytes [43]. Nrf2 over-expression or deficiency did not significantly affect NPC migration (Fig. F). Taken together, these results suggest that over-expression of Nrf2 promoted the proliferation, but not migration of isolated NPCs.

The neuronal phenotype of differentiated NPCs was determined by using Tuj-1 as a marker of neurons. After seven days of differentiation Tuj-1 positive cells were located around the neurospheres (Fig. 4A-C). In comparison to wt NPCs, the number of Tuj-1 positive neurons was about 25% higher in LV-Nrf2 NPCs ($p<0.001$) and about 30% lower in Nrf2$^{-/-}$ NPCs ($p<0.001$). When comparing LV-Nrf2 NPCs to Nrf2$^{-/-}$ NPCs the number of Tuj-1 positive neurons was about 50% higher in LV-Nrf2 NPCs ($p<0.001$) (Fig. 4D-G). Another main cell type observed were GFAP positive astrocytes (Fig. 4H). These astrocytes were mainly localized near the edge of neurospheres in close proximity to one another. Because the astrocytes did not migrate as much as neurons it was not possible to count the exact number of GFAP positive cells. To confirm the functional neuron phenotype of migrated NPCs and to test the hypothesis that Nrf2 may be involved in neuronal development, we then analyzed depolarization induced intracellular calcium $[Ca^{2+}]_i$ transients from the differentiated LV-Nrf2 and wt NPCs by using confocal Ca^{2+}-imaging. KCl treatment induces depolarization of neuronal cells, opens the voltage-gated calcium channels (VGCC) and evokes $[Ca^{2+}]_i$ transient mainly via these channels [47]. A higher proportion of LV-Nrf2 NPCs (79%) showed depolarization-induced $[Ca^{2+}]_i$ transients compared to wt NPCs (49%) ($p<0.001$), indicating differentiation to neurons (Fig. 5A-B). Furthermore, the average magnitude of KCl-induced $[Ca^{2+}]_i$ transients of LV-Nrf2 NPCs was 1.4-fold higher compared to the average magnitude of wt NPCs ($p<0.001$) (Fig. 5C-D). Finally, proliferating and differentiating wt NPCs were treated with PDTC, a potential inducer of Nrf2 and Nrf2-regulated genes [31]. Treatment with 1μM PDTC increased both the volume of neurospheres and the number of Tuj-1 positive neurons 1.2 -fold ($p<0.05$). In addition, 1 and 10 μM PDTC concentration increased also Tuj-1 positive neurons ($p<0.05$ and $p<0.01$) (Fig. 5E-F). Thus, Nrf2 enhances NPC proliferation and neuronal differentiation of NPCs, possibly upstream of VGCCs.

Nrf2 counteracts the negative effects of $A\beta_{1-42}$ on NPC viability, proliferation and neuronal differentiation

To examine the viability of NPCs in the presence of $A\beta_{1-42}$ and $A\beta_{1-40}$, differentiated NPCs were stained with SYTO 13 for live and PI for dead cells 7 days after Aβ treatment. As expected, $A\beta_{1-42}$ reduced the viability of differentiated wt and Nrf2$^{-/-}$ NPCs ($p<0.01$) (Fig. 6A). Nrf2 over-expression protected against $A\beta_{1-42}$ toxicity. $A\beta_{1-40}$ treatment had no significant effect on cell viability (data not shown). $A\beta_{1-42}$ also reduced the volume of

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neurospheres in both wt and Nrf2$^{-/-}$ NPCs by 17 and 21 % respectively, but the difference compared to control NPCs became significant only in wt cells ($p<0.05$) (Fig. 6B). Importantly, Aβ_{1-42} –induced reduction in the neurosphere volume was completely prevented in Nrf2 over-expressing NPCs. On the other hand, Aβ_{1-40} treatment reduced proliferation by almost 30 %, but only in Nrf2$^{-/-}$ NPCs ($p<0.05$). While the volume of neurospheres remained unchanged in wt NPCs, in LV-Nrf2 cells the volume of neurospheres increased almost 35 % in the presence of Aβ_{1-40} ($p<0.05$) (Fig. 6B).

Cell migration was reduced following Aβ_{1-42} treatment in all the assessed cell types. In the presence of 10 µM Aβ_{1-42}, the wt and LV-Nrf2 NPCs migrated 55 % and Nrf2$^{-/-}$ NPCs 65 % shorter distances when compared to untreated controls ($p<0.001$) (Fig. 7A). In the presence of 10 µM Aβ_{1-40} cell migration was also reduced in all NPCs groups ($p<0.001$) (Fig. 7B). Finally we studied the effect of Nrf2 on NPC differentiation in the presence of Aβ_{1-42}. Treatment with 5 µM Aβ_{1-42} did not change the number of Tuj-1 positive cells in either wt or Nrf2 over-expressing NPCs (Fig. 7C). However, treating Nrf2$^{-/-}$ NPCs with Aβ_{1-42} reduced the number of Tuj-1 positive neurons by 33 % ($p<0.05$). Taken together, these results suggest that in the presence of Aβ the over-expression of Nrf2 promotes cell viability and proliferation, but not migration of NPCs, and that expression of Nrf2 is required for normal neuronal differentiation in the presence of Aβ.

Discussion

Our findings demonstrate a crucial role for Nrf2 in injury-induced neurogenesis and shows for the first time that Nrf2 promotes NPC proliferation and neuronal differentiation. We also report for the first time that the over-expression of Nrf2 restores the NPC proliferation, differentiation and viability depressed by Aβ, a toxic peptide thought to cause synaptic dysfunction and neuronal loss in AD, the major dementing disorder. Moreover, Aβ reduced production of neurons from Nrf2-deficient NPCs. The fact that Nrf2 is protective against neurodegeneration is well established both in cell cultures of primary postmitotic neurons and in animal models [33, 35, 48], yet this study provides the first evidence of the involvement of Nrf2 in neurogenesis.

The mechanism of how Nrf2 regulates NPC proliferation and neuronal differentiation is unclear. Some studies have suggested that Nrf2 promotes proliferation and differentiation of human hematopoietic and possibly intestinal stem cells [49], whereas other studies have indicated an opposite role for Nrf2 in aortic smooth muscle cells and myofibroblasts [50-51]. Nrf2 is a master regulator of antioxidant defense mechanisms. All target genes of Nrf2 are not directly related to oxidative stress as Nrf2 can inhibit Fas-mediated apoptosis [15-16], induce expression of a class of proteosomal proteins [17-18] and also attenuate the inflammatory response [19-20]. While the mechanism of how Nrf2 over-expression promotes neuronal differentiation of NPCs remains unknown, it has been suggested that Nrf2-induced antioxidant enzymes prevent the ROS production required for stemness of the cells, thus leading to differentiation [52-54].

With respect to cellular proliferation, Nrf2 has been recently shown to promote anabolic pathways by redirecting glucose and glutamine into purine synthesis [55], which might certainly be a mechanism of Nrf2-promoted neuronal proliferation of NPCs in our study.

ROS are generally associated with increased cell proliferation, including cancer cells. Accordingly, several antioxidants reduce cell proliferation and drive differentiation. However, both clinical and experimental studies suggest that antioxidants do not prevent cancer growth but may even promote proliferation of certain cancers [56]. Nrf2 has also been shown to be over-activated in several cancers [57]. Moreover, many types of stem cells contain lower levels of ROS than their more mature progeny and this feature is considered to be critical for maintaining stem cell renewal and proliferation [58-61]. In the brain, neurogenesis declines during aging and neurodegenerative diseases, and this impairment may, at least partially, be due to increased inflammation and oxidative stress in the brain environment [1, 62-64]. Recently, aging and MPTP-induced Parkinson's disease were shown to reduce neurogenesis in the subventricular zone via impaired Nrf2-mediated adaptation to inflammation [65]. As the Nrf2 deficient mice presumably have compromised protection against oxidative stress and inflammation after transient ischemic insult, we cannot differentiate whether the blockade of ischemia-induced neurogenesis in Nrf2$^{-/-}$ mice is due to altered redox state of NPCs or more hostile brain environment with increased inflammation and release of ROS and other toxic molecules, or both. Similarly, our observation that increased Nrf2 expression enhances proliferation of cultured neurospheres might be partially due to general improvement of homeostasis and protection against challenging cell culture conditions. It is also possible that Nrf2-regulated intracellular redox state has direct effect on NPC proliferation. Importantly, recent studies have shown that Nrf2 regulates Notch signaling [66], a major regulator of proliferation of various cell types, including NPCs [67-68]. We thus hypothesize that neurogenesis might be impaired in Nrf2 deficient brain because of reduced Notch signaling. As proliferation and differentiation represent opposite phenomena and functions of cells, it is likely that the distinct Nrf2-mediated pathways allow NPCs to respond to Nrf2 activation by enhancing both proliferation and differentiation of these cells.

Nrf2 was recently reported to promote migration of glioma cells [69] and hematopoietic stem cells [49]. According to our results Nrf2 deficiency or up-regulation does not affect migration of NPCs. Because our assay measures only migration of neuronal cells, we cannot exclude the possibility that Nrf2 regulates migration of astroglial cells. Overall, together with the previous re-

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ports our findings suggest a cell-specific role of Nrf2 in regulation of proliferation, differentiation and migration.

We report here for the first time that Nrf2 over-expression mitigates $A\beta_{1-42}$ toxicity to NPCs by preventing the Aβ42-induced reduction in NPC proliferation and survival, and that Aβ42 reduces neuronal differentiation during Nrf2 deficiency, and also that Aβ40 reduces NPC proliferation during Nrf2 deficiency. These observations on the role of Nrf2 in Aβ toxicity are in agreement with previous findings showing that Nrf2 over-expression protects primary postmitotic neurons against $A\beta_{1-42}$ toxicity, improved cognitive functions in transgenic AD mice overexressing mutant APP and PS-1 genes, and that loss of Nrf2 renders postmitotic neurons more susceptible to $A\beta_{1-42}$ toxicity [27-29]. Our earlier studies [33] also showed that administration of freshly prepared $A\beta_{1-42}$ has negative effects on survival, proliferation and even migration of wt NPCs. These results indicate that Nrf2 plays a crucial role in protection of NPCs against Aβ toxicity. The fact that Nrf2 possesses both a neuroprotective and a neurogenic function raises the idea that utilizing NPCs with boosted Nrf2 for transplantation could be beneficial via several mechanisms, including alleviation of oxidative stress to endogenous brain cells, increased neurogenesis, and protection against Aβ toxicity in AD.

Conclusion

In summary, our data provides novel evidence that Nrf2 is important for regulation of neurogenesis and proliferation and neuronal differentiation of NPCs. In addition, Nrf2 over-expression has beneficial effects on survival, proliferation, and neuronal differentiation of NPCs. It is possible that NPCs in combination with increased Nrf2 activity might represent a potential therapeutic strategy to pursue in conditions such as AD, where impaired neurogenesis, Aβ toxicity and oxidative stress are important pathological features of disease.

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Discloser of Potential Conflicts of Interest

The authors indicate no potential conflicts of interest.

Author Contributions

K.V.: collection and assembly of data, data analysis and interpretation, manuscript writing, financial support; Y.P.: conception and design, collection and assembly of data, data analysis and interpretation, manuscript writing; E.S., H.D., A.K., S.L., N.N.: collection and assembly of data, data analysis and interpretation; P.T., K.M.K.: conception and design, data analysis and interpretation, manuscript writing; A.L.L., M.Y.: provision of study material, manuscript writing, critical review of the manuscript; J.M.: conception and design, data analysis and interpretation, manuscript writing; T.M.: conception and design, manuscript writing; J.K.: conception and design, financial support, administrative support, data analysis and interpretation, manuscript writing, final approval of manuscript.

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Figure 1. Nrf2 is important for endogenous neurogenesis in vivo. **(A-D):** Representative images from brain sections of wt and Nrf2$^{-/-}$ mice (Nrf2 knockout mice) subjected to global brain ischemia and stained with DCX and Ki-67 seven days after global ischemia induction to detect proliferation and differentiation of endogenous NPCs. **(E):** Endogenous neurogenesis is impaired and **(F):** cell proliferation shows a tendency to be decreased in Nrf2$^{-/-}$ mice subjected to global brain ischemia. Ki67 and DCX immunoreactivity are expressed as percentage area occupied by the immunoreactivity in DG. Bars are mean ± SEM. $^{**}P \leq 0.01$. Abbreviations: DCX, doublecortin; Ki-67, proliferation marker; DG, dentate gyrus; SGZ, subgranular zone.

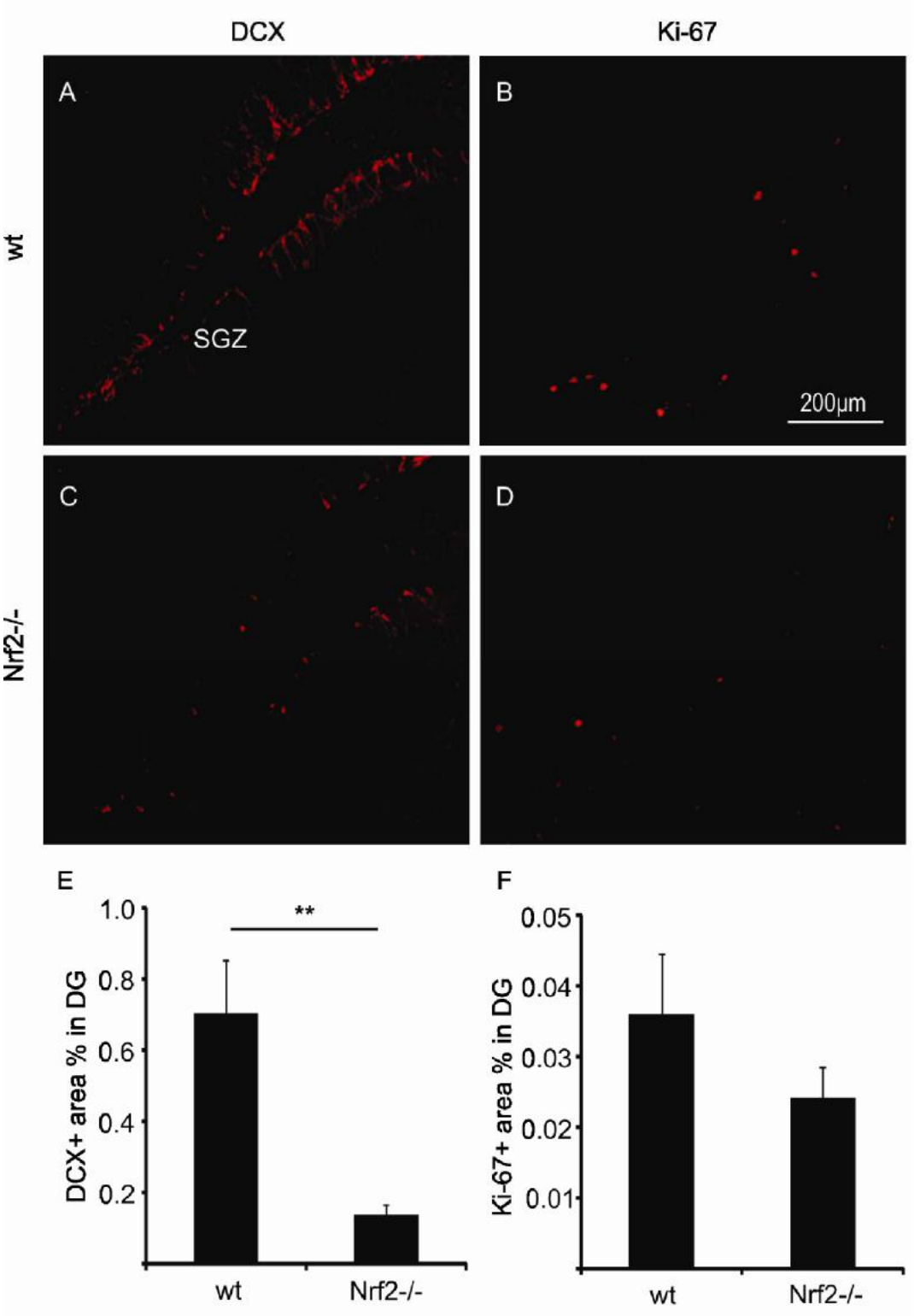

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Figure 2. Lenti-Nrf2 transduction increases expression of hNrf2 and Nrf2-regulated genes. Relative expression of mRNA encoding **(A):** hNrf2, **(B):** mHO-1, **(C):** mGCLC, **(D):** mGCLM and **(E):** mNQO1 in control and Nrf2 overexpressing NPCs (LV-Nrf2) (n = 3) were analyzed by quantitative RT-PCR. The mRNA levels are presented as mean relative expression levels normalized to β-actin. Bars are mean ± SEM. **P*≤0.05, ***P*≤0.01, ****P*≤0.001. Abbreviations: hNrf2, human Nrf2; mHO-1, mouse heme oxygenenase 1; mGCLC and mGCLM, mouse glutamate cysteine ligase and regulatory subunit; mNQO1, mouse quinone oxidoreductase 1.

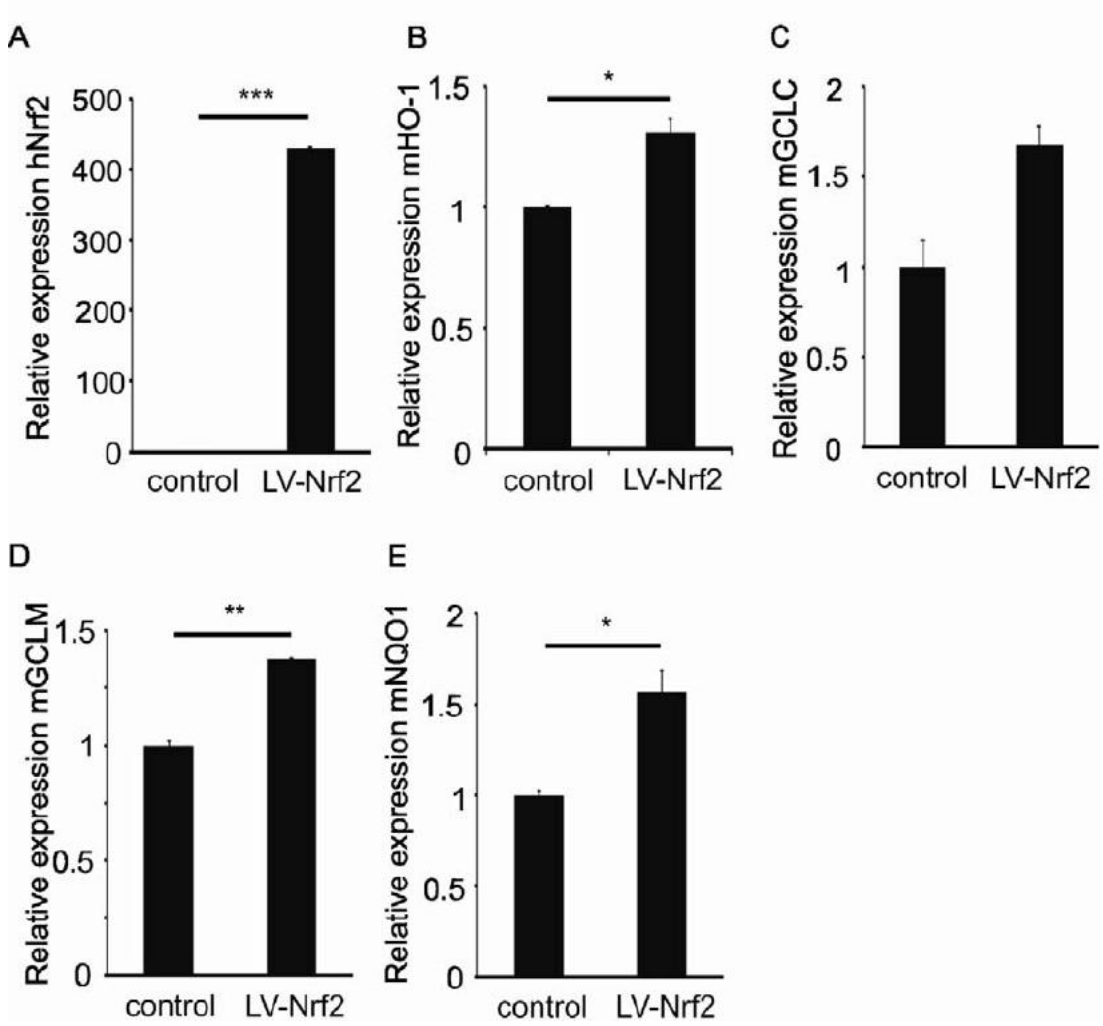

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Figure 3. Nrf2 induces proliferation but not migration of NPCs. **(A):** Representative images showing wt neurospheres, Nrf2 over-expressing neurospheres (LV-Nrf2) and Nrf2 knockout ($Nrf2^{-/-}$) neurospheres. **(B):** Proliferating NPCs were incubated in the presence of growth factors for 48 h and the volume of neurospheres was determined from 6 - 8 replicate images. Data are expressed as normalized values. **(C):** Proliferating NPCs were incubated for in the presence of 10 µM BrdU for 4 h and then immunostained. Representative images of co-staining of Hoechst (blue) and BrdU (red) wt and LV-Nrf2 NPCs. **(D):** The percentage of BrdU positive cells from 20 replicate images of wt and LV-Nrf2 NPCs were measured. **(E):** Representative images of migrating NPCs (Hoechst staining) and differentiating NPCs (GFAP and Tuj-1 staining). **(F):** Average migration distances of wt, LV-Nrf2 and $Nrf2^{-/-}$ NPCs were measured. Bars are mean ± SEM. $^{**}P \leq 0.01$, $^{***}P \leq 0.001$. Abbreviations: Tuj-1, neuronal class III β-tubulin; GFAP, glial fibrillary acidic protein; BrdU, 5-Bromo-2-deoxyuridine.

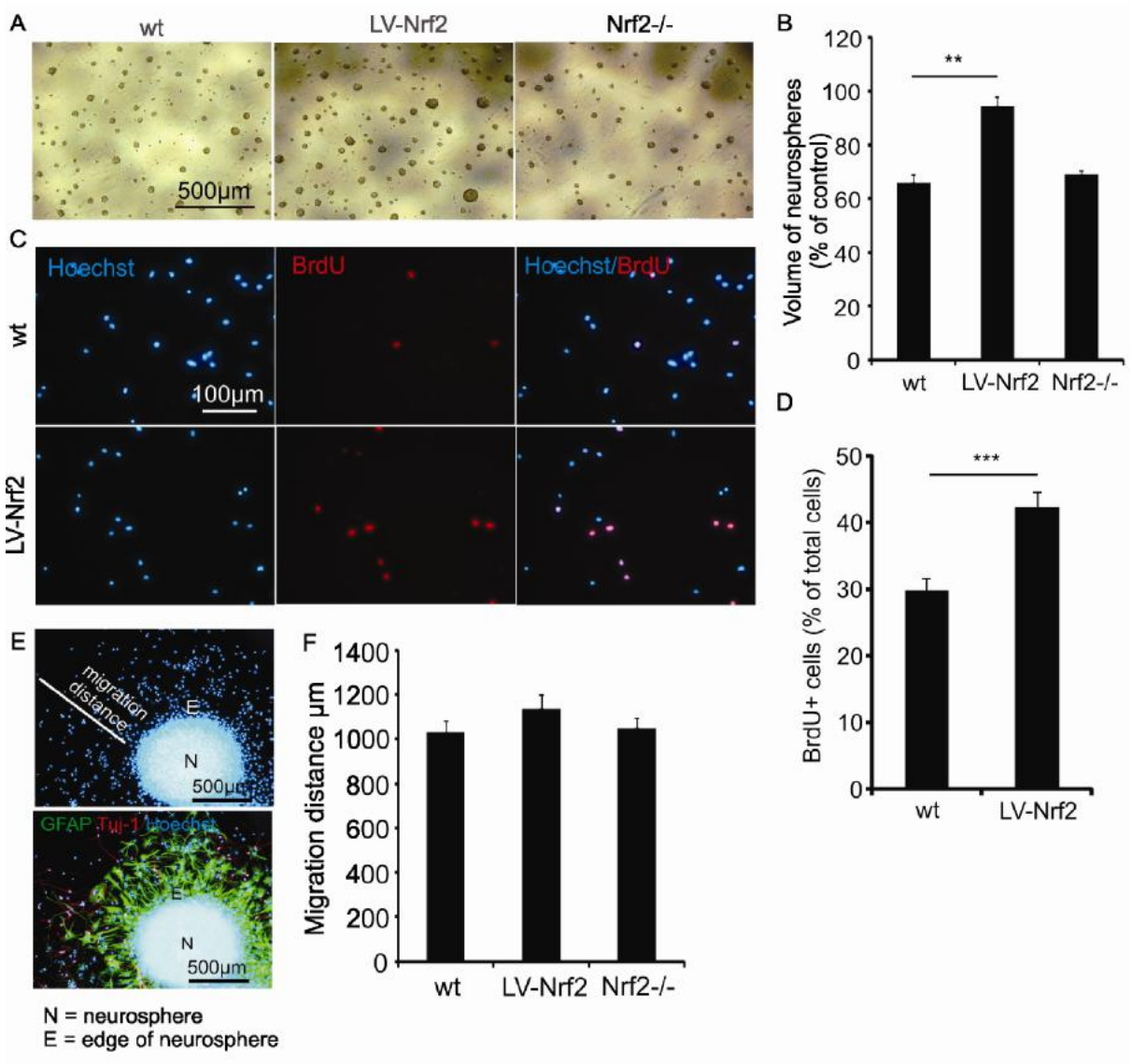

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Figure 4. Nrf2 expression regulates neuronal differentiation of NPCs. **(A-C):** The neuronal phenotype of differentiated NPCs was determined by Tuj-1 immunocytochemistry after 7 days of differentiation. Representative images of Tuj-1 positive neuronal cells in wt, Nrf2 over-expressing (LV-Nrf2) and Nrf2 knockout ($Nrf2^{-/-}$) cells. **(D-F):** Co-staining of Tuj-1 positive neuronal cells (red) with nuclear Hoechst stain (blue) in wt, LV-Nrf2 and $Nrf2^{-/-}$ cells. **(G):** The percentage of Tuj-1 positive cells from 10 - 12 replicate images was determined from the periphery of the neurosphere colony area where the NPCs grow separately. Tuj-1 immunoreactivity is expressed as the percentage of viable cells. **(H):** A representative image of GFAP positive astrocytes (green) which located near the edge of the neurosphere in close proximity to one another. Bars are mean ± SEM. $^{**}P \leq 0.01$, $^{***}P \leq 0.001$. Abbreviations: Tuj-1, neuronal class III β-tubulin; GFAP, glial fibrillary acidic protein.

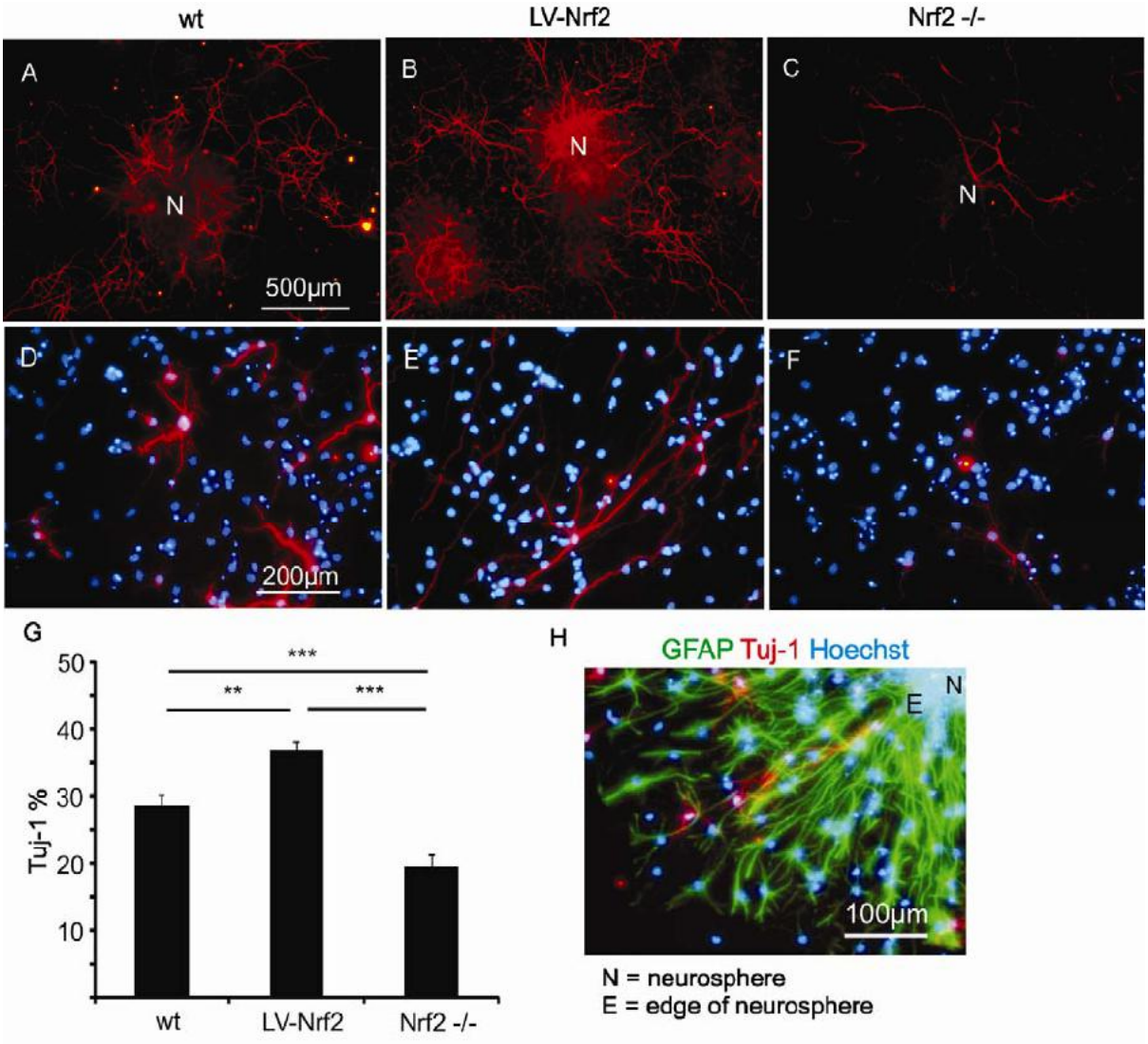

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Figure 5. Nrf2 over-expression promotes neuronal differentiation of NPCs determined with Ca2+ imaging and pharmacological activation of Nrf2 increases proliferation and neuronal differentiation of NPCs. The percentage of cells displaying a neuronal phenotype was determined in wt and Nrf2 over-expressing NPCs based on their responsiveness to depolarization. **(A):** After 6 days of differentiation a higher proportion of LV-Nrf2 NPCs produced KCl-induced Ca^{2+} transients compared to wt NPCs. **(B):** Representative images of cells before and after KCl stimulation. **(C):** The amplitude of F/F0 $[Ca^{2+}]_i$ of wt (n=80) and LV-Nrf2 (n=88) NPCs to KCl-induced Ca^{2+} transient. **(D):** Representative images of KCl-induced Ca^{2+} transient from wt and LV-Nrf2 NPCs. **(E):** The volume of neurospheres 48 h after PDTC treatment. The average volume of the largest neurospheres (>40µm) was counted from 6 – 8 replicates. Data are expressed as normalized values. **(F):** The percentage of Tuj-1 positive neuronal cells in PDTC treated wt NPCs after 7 days differentiation was determined from 10 to 12 replicate images by using Tuj-1 immunocytochemistry. Results are normalized to untreated control cells. Bars are mean ± SEM. **P*≤0.05, ***P*≤0.01, ****P*≤0.001. Abbreviations: KCl, potassium chloride; Ca^{2+}, calcium; F/F0, fluorescence intensity; PDTC, pyrrolidine ditiocarbamate.

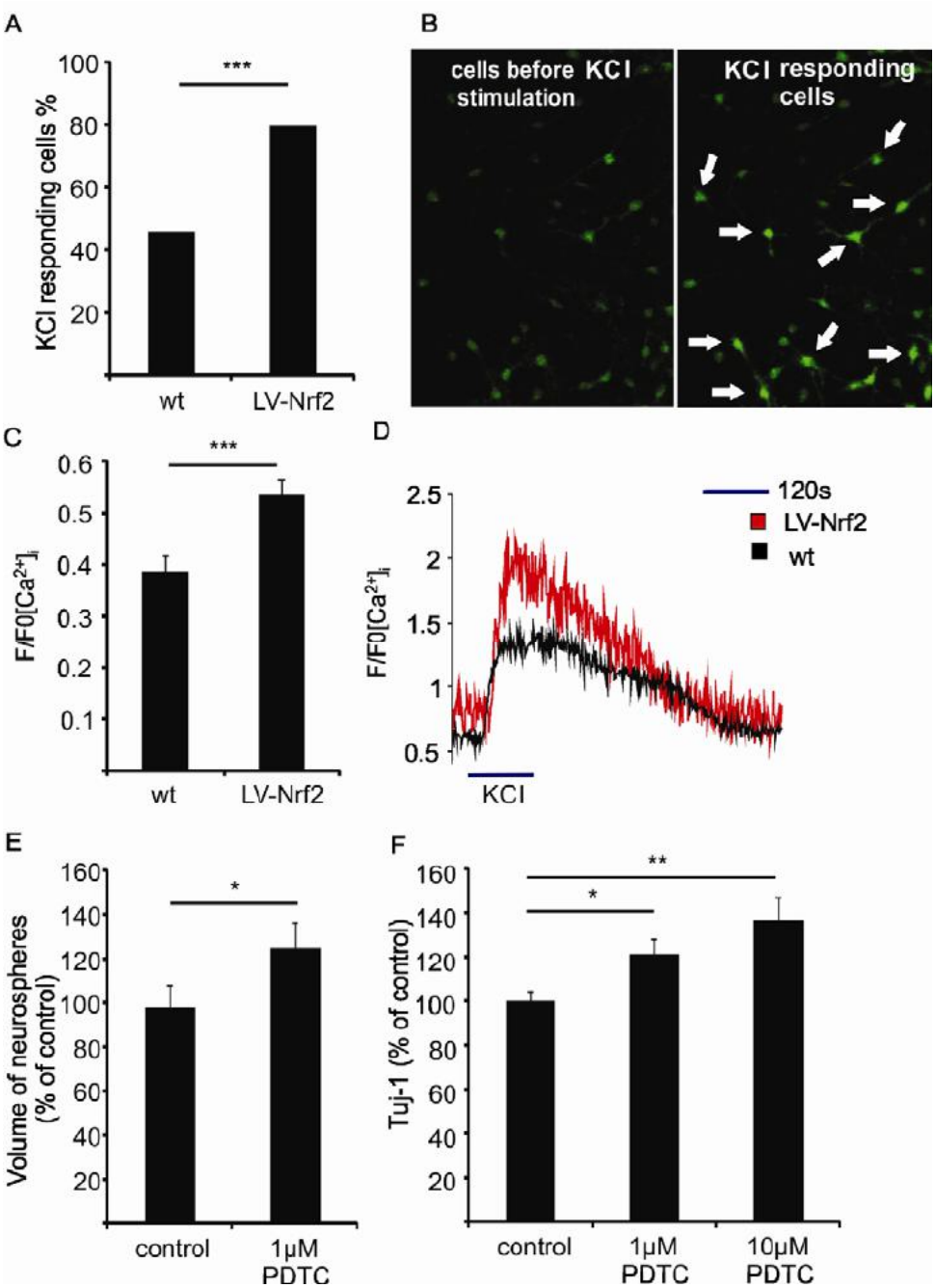

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Figure 6. Nrf2 counteracts the negative effects of $A\beta_{1-42}$ on NPC viability and proliferation. **(A):** Differentiated wt, Nrf2 over-expressing (LV-Nrf2) and Nrf2 knockout ($Nrf2^{-/-}$) NPCs were treated for 7 days with 2.5 µM $A\beta_{1-42}$. Cell viability was assessed by counting the number of cells from 6-8 replicates images following SYTO 13 and PI staining. **(B):** Proliferating NPCs were treated with 30 µM $A\beta_{1-42}$ or $A\beta_{1-40}$ for 48 h. The average volume of the largest neurospheres (>40µm) was counted from 6-8 replicate images. Results are normalized to untreated control cells. Bars are mean ± SEM. **P*≤0.05, ***P*≤0.01. Abbreviations: Aβ, amyloid beta; SYTO 13, marker for live cells; PI, probidium iodide.

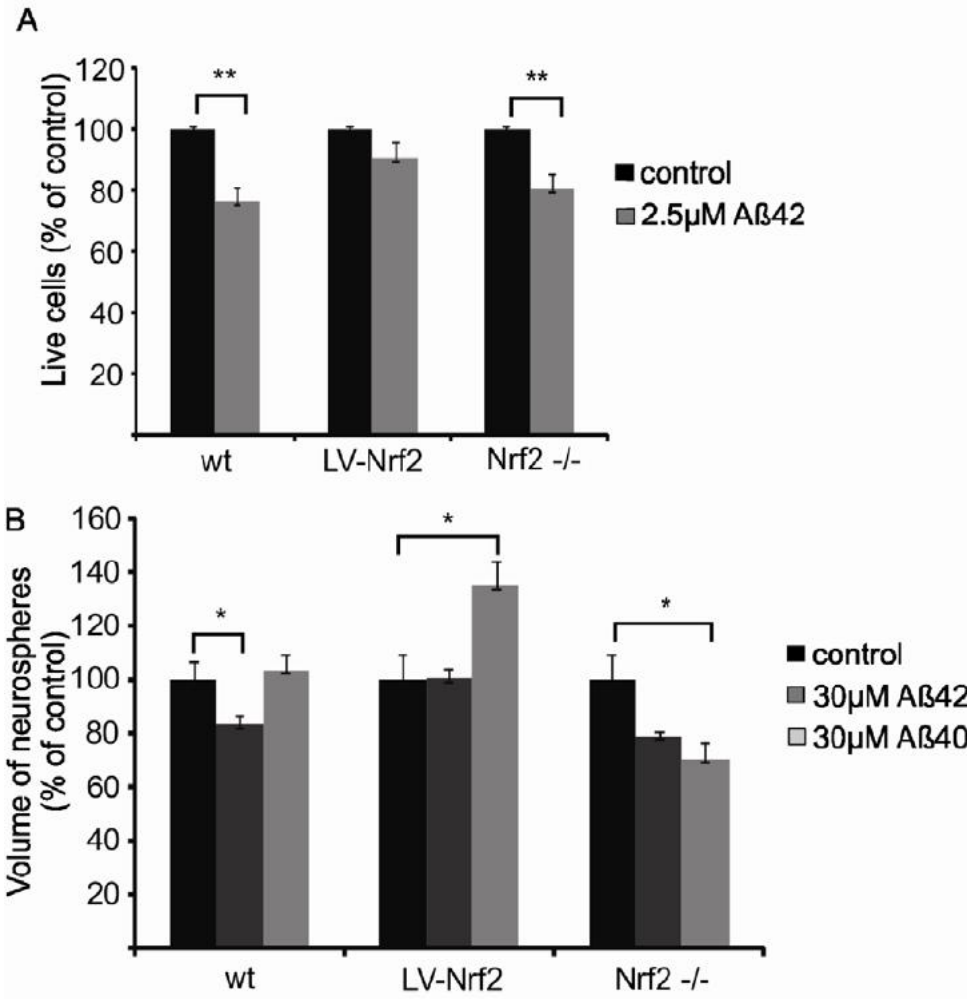

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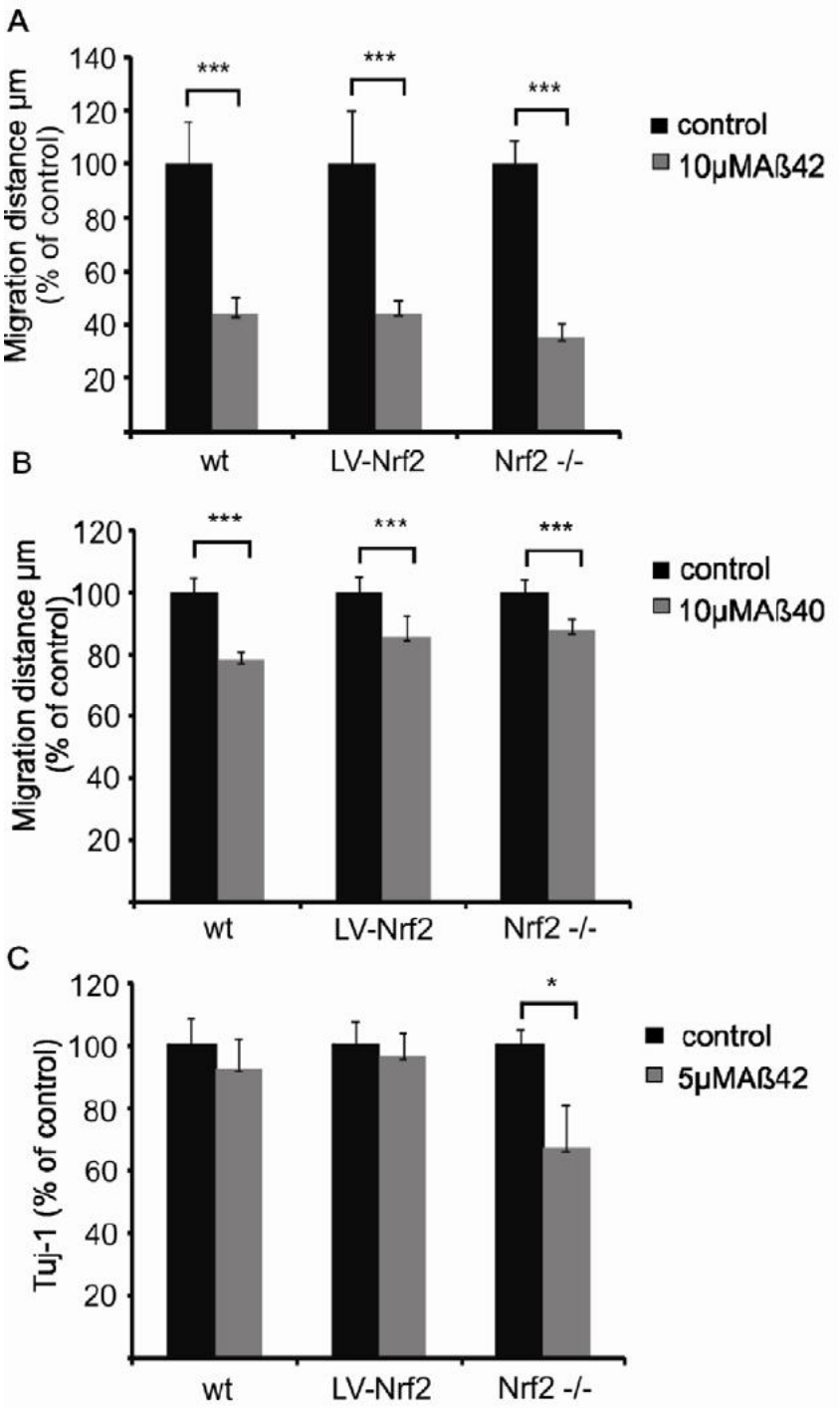
Figure 7. Nrf2 deficiency impairs neuronal differentiation but not migration of NPCs in the presence of Aβ. **(A-B):** Differentiated wt, Nrf2 over-expressing (LV-Nrf2) and Nrf2 knockout ($Nrf2^{-/-}$) NPCs were treated with $A\beta_{1-42}$ and $A\beta_{1-40}$. The average migration distance (µm) was measured from 6 – 8 replicate images from the edge of the neurospheres. The data are normalized to untreated control cells. **(C):** Treatment of NPCs with $A\beta_{1-42}$ for 7 days reduced neuronal differentiation in $Nrf2^{-/-}$ cells as determined from 10 to 12 replicate images by Tuj-1 immunocytochemistry. Tuj-1 immunoreactivity is normalized to untreated control cells. Bars are mean ± SEM. *P≤0.05, ***P≤0.001. Abbreviations: Aβ, amyloid beta; neuronal class III β-tubulin.



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